Class Test 1 – Parametrization of Curves

Multivariable Calculus

Time: 40 minutes **Max Marks:** 20

Maintain academic integrity. Cheating will not be tolerated

1. Find a parametrization of the line segment joining $(2, 1, 1)$ and $(5, 4, 3)$.

2. Given
$$x = 3\cos t, \quad y = 3\sin t,$$
(i) Eliminate the parameter. (ii) Identify the curve.

3. Consider
$$x = \cos t, \quad y = \sin t, \quad 0 \leq t \leq 2\pi.$$
(i) Is the curve traced clockwise or counterclockwise as t increases?
(ii) Find a parametrization of the same curve that traces it in the opposite direction.

4. Show that the curves
$$x = \cos t,\ y = \sin t \quad \text{and} \quad x = \cos(2s),\ y = \sin(2s)$$
represent the same circle. How does the tracing differ?